Prisoners Dilemma Competition

(Thanks to David Parkes from Harvard University for this question). In his 1984 book *The Evolution of Cooperation*, Robert Axelrod described the results of a tournament he organized for academics to submit strategies for playing the prisoners dilemma for a fixed number of steps. The winner of Axelrod's competition was the strategy "tit for tat" – start by cooperating, and simply choose the same move as was chosen by the opponent in the previous round. In general, Axelrod found that altruistic strategies triumphed over greedy ones.

For this problem set, you will create a submission to a tournament like the one Axelrod set up in 1984 but with a twist. The information your player receives about its opponent's previous action is noisy, i.e., with probability 0.05 your player receives incorrect information about the previous action of the other player. What do you think would be the best strategy to overcome a noisy environment for the iterated prisoners' dilemma? Write an agent for playing the prisoners' dilemma in Python. Your agent needs to specify whether to play "C" or "D" depending on its observations, which consist of the history of play. Each match is 100 rounds, and every agent plays every other agent.

Use the support code (`student_pack.zip`): write your agent in `my_agent.py` and test it with `python3 practice.py`. The `README.md` explains how.

- Do you think Tit-for-Tat is a good overall strategy for the noisy domain?
- Write a few paragraphs (in English or Hebrew) describing your agent's strategy. Evaluate the agent empirically.
- Submit one zip file named with your ID number, `ID.zip` (e.g. `123456789.zip`), containing your agent `my_agent.py` and your answers `report.pdf`.
- The first place winner will get a 1 pt bonus in the course.

I will conduct a tournament between all submissions using the support code. Note that you cannot get credit for this assignment if your agent does not run with `practice.py` on a standard Python 3 installation or needs any external packages.